DEMONSTRATION ON OPEN DATA · NOVEMBER 2026

Monitoring note: Cleve Hill Solar Park

What satellite data shows between survey visits: construction, the habitat management area, and what to check next

SITE	Cleve Hill Solar Park (EN010085), Graveney Marshes, Kent; next to The Swale SSSI, SPA and Ramsar site
COVERED	January 2017 to October 2026: 492 clear Sentinel-2 views and 1,910 Sentinel-1 passes
REFERENCE	Outline Landscape and Biodiversity Management Plan, Rev E (November 2019); project newsletter (June 2024); Quinbrook (1 July 2025)
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Status. A demonstration on open data, not client work. Not a site visit and not a compliance assessment: satellite data shows where and when something changed; the ecologists and the Habitat Management Steering Group judge what it means. Areas are approximations from open data.

Summary

- **Construction, dated:** half the mapped arrays had steel by June 2024 and modules by October 2024. On 8 June 2024 Sentinel-2 saw modules on 7.3% of the arrays and radar saw steel on 51%, against 7.4% of modules and 57.4% of piles in the project's newsletter.
- **Arrays:** 202 ha mapped from Sentinel-2, covering 96% of the panel blocks drawn in OpenStreetMap.
- **Habitat management area:** about 56 ha of arable land east of the compound (the plan: c. 56 ha), cultivated and sown in August and September 2023, green from late October 2023 and through the first winter of construction, as the plan intended.
- **The plan's 50 m exclusions** (modules and transformers, flood defence crest, road edges) take 1.7 ha of it, 3%.
- **Management signals:** sharp falls in greenness on 29 May 2025 (on 78% of the area); 6 June 2026 (on 38% of the area); none in August to October, before winter.
- **Water:** no more than 0.4 ha of open water on any radar pass in the two winters since the wetland works, and none on clear Sentinel-2 views.

To check on the next visit

- **Sward height before winter** (the plan: under 100 mm). No cut or sharp fall showed from August to October in 2024, 2025 or 2026, and grazing at the plan's densities doesn't show from space.
- **The falls of late May 2025 and early June 2026:** a cut, heavy grazing or quick browning? Both springs were dry (-31% and -38% rain from March to August). How do they fit the breeding season of ground-nesting birds?
- **Bare ground mapped in September and October 2026** (3.2 ha, figure 2): works, poaching, reseeding or drought?
- **Water in ditches and scrapes:** neither sensor saw standing water on the area in the last two winters. Are the water control measures holding water as intended?

1 Construction, dated



Figure 1. When modules first showed on each part of the arrays (Sentinel-2), and the share of the arrays with steel (Sentinel-1) and modules over time.

- **Arrays:** modules are dark in near infrared and bright in shortwave infrared, so they stand apart from grass, crops and soil. 202 ha mapped; with 550,000 modules of a typical size (about 142 ha of glass), that is a ground cover of about 70%, as expected for east-west arrays.
- **Two steps in the radar record:** the steel frames raise the radar return by 3 to 5 dB; the modules then send it away, 4 to 5 dB below the level before construction. Modules went on a median of 137 days after the steel.
- **Against the published milestones** (8 June 2024): modules on 7.3% of the arrays (newsletter: 7.4% of modules); steel on 51% (57.4% of piles). 95% of the arrays had modules by February 2025, before commissioning in May 2025.

2 The habitat management area

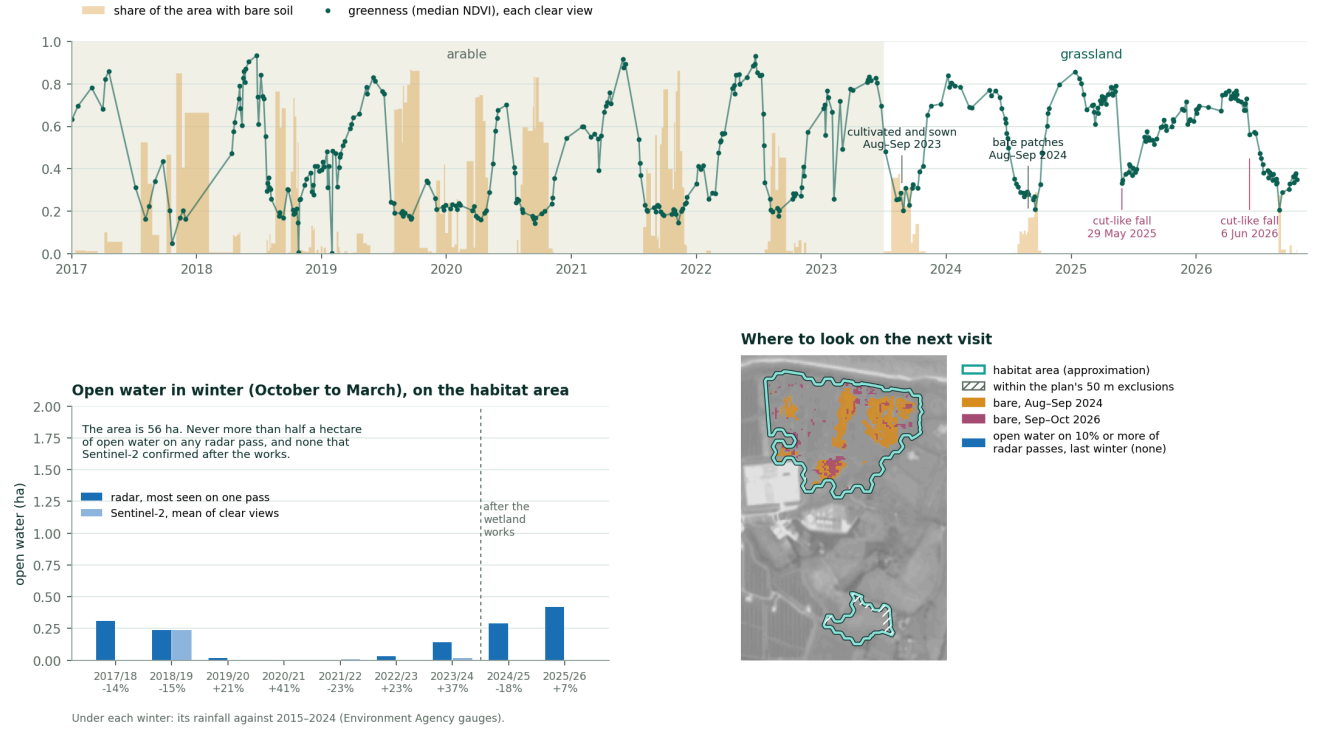


Figure 2. The habitat management area: greenness and bare soil since 2017, open water each winter, and where to look on the next visit.

- **The area:** land in the Order limits east of the compound that the Crop Map of England shows as arable in 2020 to 2022 and as grass in 2024: 55.6 ha. The agreed boundary is the steering group's; this is an approximation for monitoring.
- **From arable to grassland:** the last crop came off in July 2023; bare soil from August to mid-September 2023; green from late October. Every winter before showed bare soil; none since.
- **After the change:** bare patches in August and September 2024 (11.9 ha), which fit the scrape and swale works; almost none in 2025 (0.1 ha); 3.2 ha in September and October 2026.

3 What satellite data can't tell here

- **Sward height and structure,** the plan's main target for the wintering birds. Greenness is not height.
- **Grazing** at a few sheep per hectare, which changes the sward too gradually to show as an event.
- **Bird use,** which needs the winter counts; and species and condition, which need the survey.
- **The ground under the modules:** at about 70% cover, Sentinel-2 sees the modules, not the grass between and beneath them.
- **Small or short-lived water:** scrapes smaller than a few 10 m pixels, or flooding that drains between passes.

4 Method and data

Sentinel-2 L2A (Earth Search), every view with at least 30% of the area clear, 2017 to 2026; Sentinel-1 RTC (Planetary Computer), every pass, VV and VH. Arrays: median NDMI below -0.30 since operation. Module date: first clear view from which NDMI stays below -0.30 in 80% of later views. Steel: VV 3 dB above the 2021 to 2023 level on three passes running, on 100 m cells. Management signals: NDVI falls over 0.2 between clear views on at least a quarter of the area (Kolecka et al. 2018). Bare soil: BSI over 0.05 and NDVI under 0.2. Water: VV under -18 dB and VH under -23 dB, frost and snow passes skipped (ERA5-Land), Sentinel-2 MNDWI as referee. Context: Order limits (Planning Data), The Swale (Natural England), Crop Map of England 2020 to 2024 (RPA), flood defences and rain gauges (Environment Agency), all under the Open Government Licence; panel blocks from OpenStreetMap (ODbL) as a check only.

5 How AI was used

An agent (Claude Code) wrote the scripts: downloads, the array map, the dating, the habitat record, the figures and this note. Every result was checked against something independent: OpenStreetMap's panel blocks, the project's published progress figures and dates, the Crop Map of England and the plan's area. What it got wrong first: it expected radar to brighten over the arrays (it darkens once the modules are on), dated construction pixel by pixel (radar speckle pushed dates months late; 100 m cells fixed it), drew the habitat area from all unfenced land (93 ha, until the crop map narrowed it to the arable reversion), and counted snow as water in early 2018.

6 Sources

- Cleve Hill Solar Park Ltd (2019). Outline Landscape and Biodiversity Management Plan, Rev E, EN010085-001707. Community newsletter, June 2024. Quinbrook, 1 July 2025.
- Kolecka, N. et al. (2018). Regional scale mapping of grassland mowing frequency with Sentinel-2 time series. Remote Sensing 10, 1221. Reinermann, S. et al. (2022). Detection of grassland mowing events for Germany by combining Sentinel-1 and Sentinel-2 time series. Remote Sensing 14, 1647.

Contains modified Copernicus Sentinel data (2017 to 2026) and ERA5-Land information (C3S). Contains public sector information licensed under the Open Government Licence v3.0 (Planning Inspectorate, Natural England, Rural Payments Agency, Environment Agency). © OpenStreetMap contributors (ODbL).